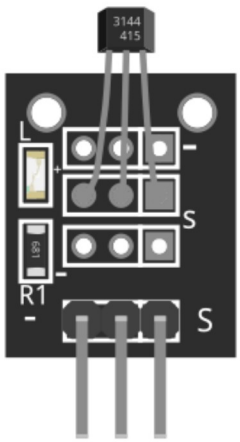
KY-003 HALL MAGNETIC SENSOR MODULE

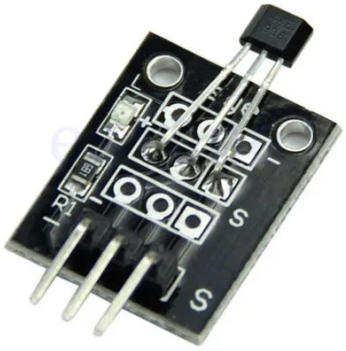
February 25, 2023 by ArduinoModules

Magnetic, Switches & Input

The KY-003 Hall Magnetic Sensor module is a switch that reacts to the presence of a magnetic field, turning itself on or off. Compatible with popular microcontrolers like Arduino, Raspberry Pi and ESP32.

This module offers a digital output, it looks similar to the KY-035 analog hall magnetic sensor, and it is functionally similar to the KY-024, a digital/analog magnetic sensor.





KY-003 SPECIFICATIONS

This module consists of a 3144EUA-S Hall-effect switch, a 680Ω resistor, an LED and 3 male header pins.

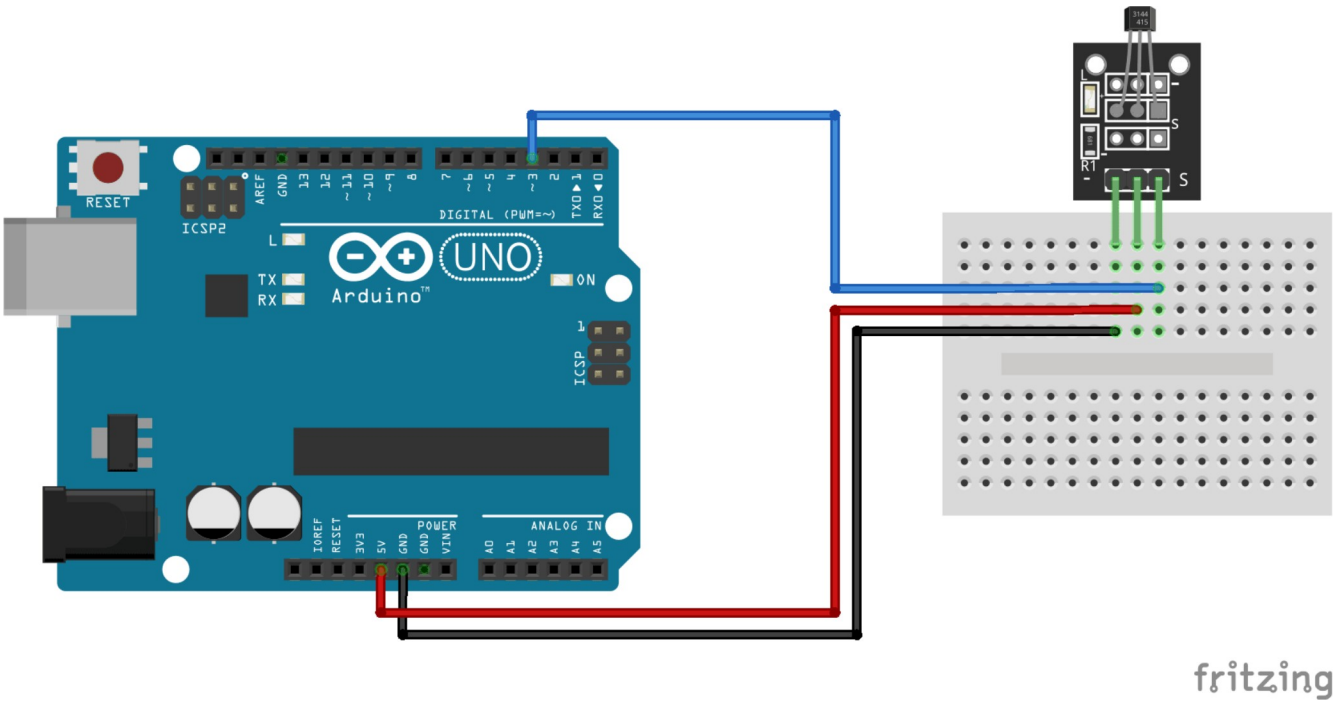
Operating Voltage	4.5V to 24V
Operating Temperature Range	-40°C to 85°C [-40°F to 185°F]
Board Dimensions	18.5mm x 15mm [0.728in x 0.591in]

CONNECTION DIAGRAM

Connect the board power line (middle) and ground (-) to +5 and GND on the Arduino respectively.

Connect signal (S) to pin 3 on the Arduino.

KY-003	Arduino
S	Pin 3
middle	+5V
–	GND



KY-003 ARDUINO CODE

The following Arduino sketch will turn on the LED on pin 13 when you put a magnet near the module.

```
int led = 13;//LED pin
int sensor = 3; //sensor pin
int val; //numeric variable

void setup()
{
    pinMode(led, OUTPUT); //set LED pin as output
    pinMode(sensor, INPUT); //set sensor pin as input
}

void loop()
{
    val = digitalRead(sensor); //Read the sensor
    if(val == LOW) //when magnetic field is detected, turn led on
    {
```

```
        digitalWrite(led, HIGH);
    }
    else
    {
        digitalWrite(led, LOW);
    }
}
```

DOWNLOADS

- Fritzing Part: KY-003 Hall Magnetic Sensor Module.
- 3144 Datasheet by Allegro Microsytems.
- Fritzing Official Site.

Magnetic, Switches & Input

Arduino, Hall Magnetic, Keyes, KY-003, Magnetic

KY-010 Photo Interrupter Module

KY-006 Passive Buzzer Module

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Simba manyeredzi

5 years ago

I like to have a signal sensor

Reply

Pingback: Sensori magnetici: reed vs. effetto Hall | scalaeNNe - Note Sparse (Treni, Ferrovie e loro modellazione in Scala N)

mattia

6 years ago

guaglio ma come cazzo si feic la simulazione

Reply

Ilaria chiodi

6 years ago

Mattia ricorda di portare quel foglio

Reply

Gisele Gotardi

7 years ago

Is the output of this sensor digital or analog? I saw a video of a guy who tested the ky-003 and the serial plotter showed a linear output according to the strength of the magnetic field. However, I always find in the descriptions of the ky-003 that it is a digital sensor. Could you help me with this clarification?

Reply

marc

5 years ago

I have a ky-003 here Gisele and per its datasheet (there's a Schmitt trigger there) it does appear to hard switch. There are linear sensors but the one on the ky-003 is a solid-state switch by my own tests. The block diagram suggests the trigger is driving the base of an open collector NPN transistor so it can source current directly.

I just checked with one I have here and there's a very clear hysteresis effect from the Schmitt trigger as so there's a small "dead spot" (designed) so the device doesn't stutter as the magnetic passes through the sensitive range.

What you're thinking of is the KY-024 which is slightly more complex on board but has both digital and analog outputs.

Reply

simon

8 years ago

this line should be… 14 if(val == LOW)

Reply

ArduinoModules

8 years ago

You're right, thanks! I updated the code.

Reply

Jeffrey

9 years ago

Amazing dood

Reply

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Frank Allen on KY-028 Digital Temperature Sensor Module

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